Experiment 18: Experiment to understand Matplotlib library use cases in Data Science through visualization

Code:

```
import pandas as pd

import numpy as np

import seaborn as sns

import matplotlib.pyplot as plt


data = pd.read_csv('/content/Dataset.csv')


data.info()

data.describe()

data.value_counts('variety')


sns.countplot(x='variety', data=data)

plt.show()


dummies = pd.get_dummies(data.variety)

FinalDataset = pd.concat([pd.get_dummies(data.variety), data.iloc[:, [0, 1, 2, 3]]], axis=1)

FinalDataset.head()


sns.pairplot(data, hue='variety', height=2.5)

plt.show()
```

```
sns.FacetGrid(data, hue='variety', height=5).map(sns.histplot,
'petallength(cm)').add_legend()

plt.show()


sns.FacetGrid(data, hue='variety', height=5).map(sns.histplot,
'PetalWidth(cm)').add_legend()

plt.show()


sns.FacetGrid(data, hue='variety', height=5).map(sns.histplot,
'SepalLength(cm)').add_legend()

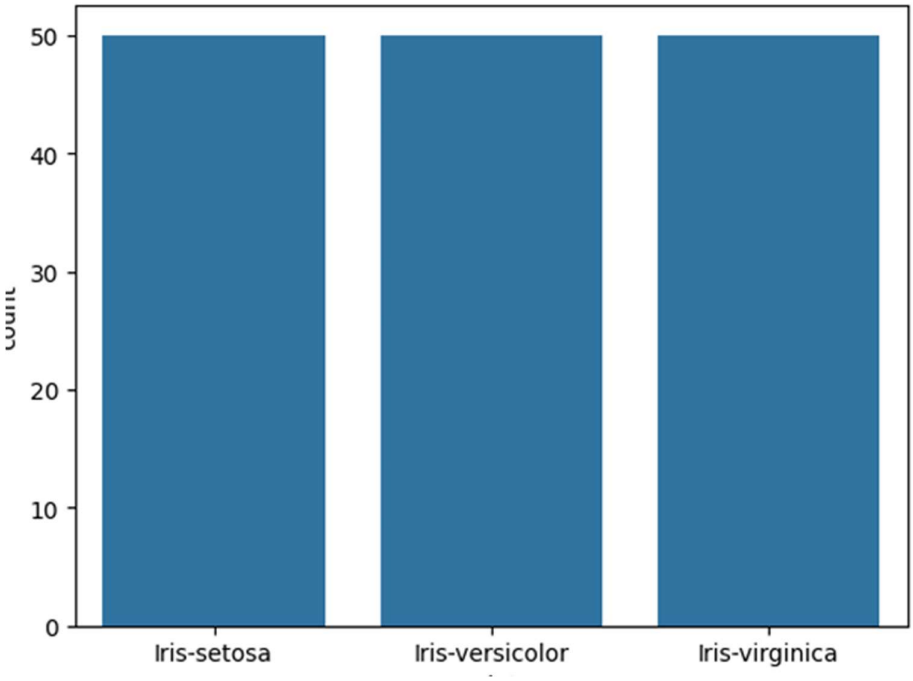
plt.show()

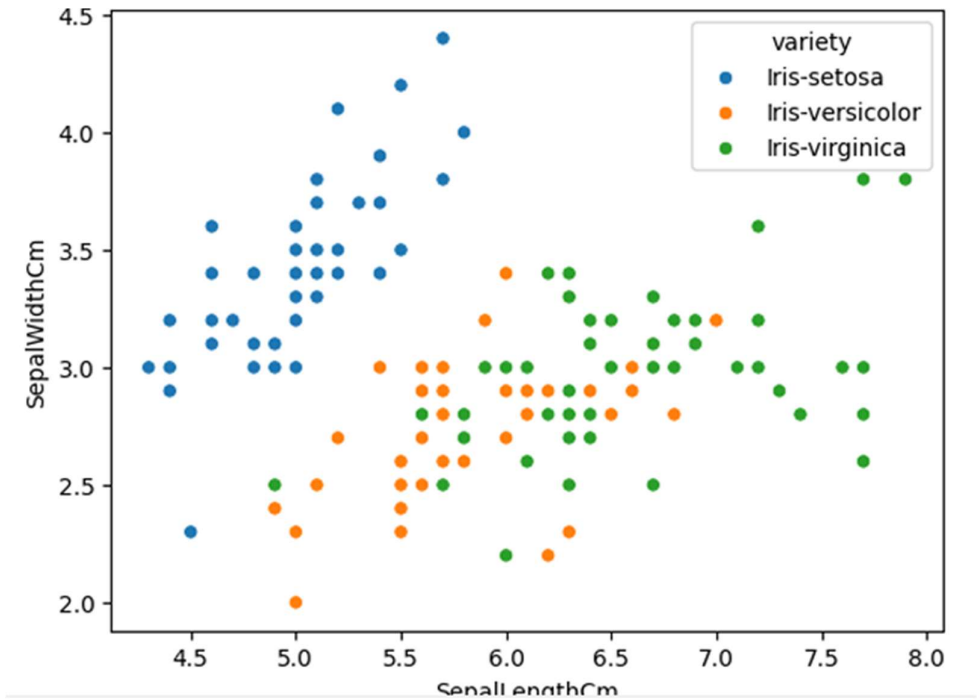

sns.FacetGrid(data, hue='variety', height=5).map(sns.histplot,
'SepalWidth(cm)').add_legend()

plt.show()
```

Output:

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm
count	150.000000	150.000000	150.000000	150.000000	150.000000
mean	75.500000	5.843333	3.054000	3.758667	1.198667
std	43.445368	0.828066	0.433594	1.764420	0.763161
min	1.000000	4.300000	2.000000	1.000000	0.100000
25%	38.250000	5.100000	2.800000	1.600000	0.300000
50%	75.500000	5.800000	3.000000	4.350000	1.300000
75%	112.750000	6.400000	3.300000	5.100000	1.800000





Result:

Thus , the Matplotlib library use cases in Data Science through visualization was analysed and executed successfully.